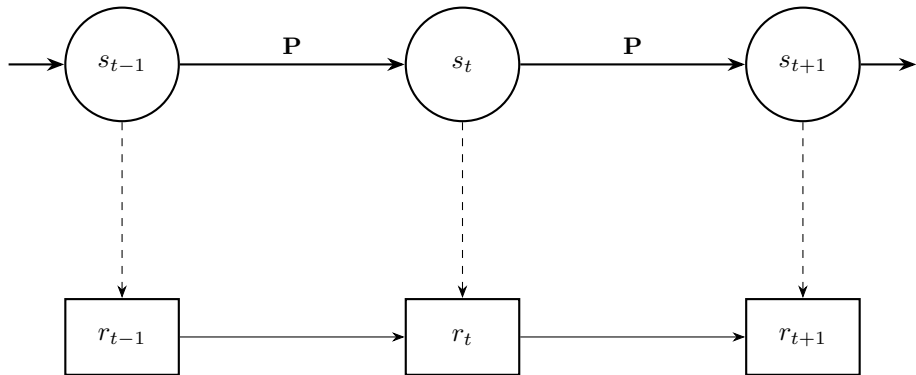




$$r_t = \varphi_{s_t} r_{t-1} + \varepsilon_t, \quad \varepsilon_t \mid s_t \sim \mathcal{N}(0, \sigma_{s_t}^2)$$